DeepSeek vs. GPT: A Comparative Analysis of AI-Generated Resumes for Job Description

1st Venkata Sai Kumar Rachagundla
Anuradha and Vikas Sinha Department of Data Science
University of North Texas
Denton, USA
venkatasaikumarrachagundla@my.unt.edu

2nd Harish Chinnakadiri
Anuradha and Vikas Sinha Department of Data Science
University of North Texas
Denton, USA
harishchinnakadiri@my.unt.edu

3rd Nivas Kumar Rao Aineni
Anuradha and Vikas Sinha Department of Data Science
University of North Texas
Denton, USA
nivaskumarraoaineni@my.unt.edu

***Abstract*—As AI recruitment increases in prominence, resume production using Large Language Models (LLMs) has picked up momentum. In the present project, DeepSeek vs. GPT: A Comparative Examination of AI-Generated Resumes for Job Descriptions, the efficiency of LLMs—GPT, in this case—DeepSeek R1 in the creation of Applicant Tracking System (ATS)-friendly resumes is assessed. Resumes are produced using several types of prompts as input, namely zero-shot, few-shot, and chain-of-thought, using job descriptions obtained through websites like LinkedIn, Dice, and Indeed. These resumes are tested using SkillSyncer software that works as an ATS to detect the alignment of the resume in terms of keywords, format, and readability. The project aims to determine which LLM performs the best in many industry sectors such as Healthcare, Technology, and Finance. It adopts a hypothesis-driven approach by comparing resume quality among models and whether DeepSeek models perform better in a significant way than GPT models. This work bridges AI and recruitment through data-driven evidence and actionable recommendations on the best models and prompting practice for resume building and ultimately improving job-seeking results in a more automated recruitment world.**

***Index Terms*—Large Language Models (LLMs), Resume Generation, Prompt Engineering, Applicant Tracking Systems (ATS), DeepSeek vs. GPT, SkillSyncer (ATS Tool), Job Description Alignment, AI in Recruitment, Zero-shot, Few-shot, Chain-of-Thought Prompting, Comparative Analysis**

I. Introduction

The integration of Artificial Intelligence (AI) into modern recruitment workflows has fundamentally reshaped the traditional methods of resume creation, evaluation, and candidate shortlisting. Among these innovations, Large Language Models (LLMs) such as OpenAI's GPT and DeepSeek R1 have emerged as transformative technologies, capable of generating highly contextual and personalized resumes with minimal human intervention. These models utilize sophisticated natural language understanding and generation capabilities to align a candidate's profile with job descriptions, often drawn directly from postings on professional platforms like LinkedIn, Dice, and Indeed.

Despite their growing adoption among job seekers, a significant question remains largely unanswered: How effective are these AI-generated resumes in passing through automated recruitment systems like Applicant Tracking Systems (ATS)? While these systems are widely employed by employers to screen and filter resumes, many AI-generated documents may fail to meet the underlying structural, lexical, and semantic criteria required to score well within such platforms. This presents a crucial challenge: without empirical validation, the utility of LLMs in resume generation remains speculative rather than substantiated.

This research addresses a critical gap in the field—the comparative evaluation of different LLMs in generating resumes that are optimized for ATS compliance. Prior studies have demonstrated the power of transformer-based models in capturing contextual semantics and improving natural language tasks; however, few have rigorously benchmarked these models against real-world tools used in recruitment pipelines. Moreover, the effectiveness of various prompt engineering techniques, including zero-shot, few-shot, and chain-of-thought prompting, remains underexplored in the context of resume writing. These prompt strategies significantly influence the structure, relevance, and quality of AI-generated content, thus warranting a systematic investigation.

To this end, the present study implements a structured experimental framework. Job descriptions are collected from leading job portals using web scraping and manual curation methods to ensure industry diversity and content accuracy. These descriptions are used as inputs for LLMs through standardized prompt templates, enabling consistent genera-

tion of AI-driven resumes across multiple domains such as Healthcare, Technology, and Finance. The generated resumes are then evaluated using SkillSyncer, a recognized ATS tool that assesses keyword alignment, formatting compliance, and readability metrics to simulate recruiter filtering.

The core objective of this study is to identify which LLM—among GPT, DeepSeek R1, and others—combined with the appropriate prompting technique, yields the most ATS-optimized resumes. Furthermore, the study seeks to offer evidence-based insights and actionable recommendations for job seekers, HR professionals, and AI tool developers. By bridging the gap between generative AI capabilities and recruitment system expectations, this work contributes to the advancement of AI-enabled hiring technologies and supports the development of more intelligent, adaptive, and effective job application tools.

II. Review of Literature

Prompt engineering is now an accepted way to interface with large language models (LLMs) toward multiple functions in medicine, education, AI product development, resumes, and computer graphics. The literature discusses the possibility of prompting an AI response for performance, relevance, and human alignment.

In health, [1] have made a case for prompt engineering playing a growing role in general practice and patient instructions and notes. Discussed by [2] is the DeepSeek-R1 open-source LLM that, through multistage reinforcement learning together with mixture-of-experts (MoE) architecture, gathers real-time diagnostic reasoning-all at a reduced-resource cost of the overall scheme. [3] benchmarked DeepSeek against Chat GPT and LLaMA for clinical tasks where DeepSeek was on par with diagnostic reasoning but slightly lagged behind in report summarization quality.

In education, [4] evidenced the few-shot prompting effectiveness for question generation in English learning, where a few examples were enough to support ChatGPT in articulating statistically valid questions. [5] looked into students' use of ChatGPT for writing assignments, revealing how iterative prompting enhanced both content personalization and clarity.

[6] presented Resume2Vec, a framework that used transformer-based deep learning models to boost applicant tracking systems (ATS). Typical traditional ATS systems depend upon keyword filtering, which fail to understand the contextual meaning in resumes and job descriptions; Resume2Vec is a solution to this problem. It uses embeddings from encoders like BERT, RoBERTa, and DistilBERT, and decoders like GPT, Gemini, and Llama. Resume2Vec uses cosine similarity to compare the alignment between the resumes and the job descriptions, and in this way, the framework improves upon traditional ATS approaches by 15.85 % in Normalized Discounted Cumulative Gain (nDCG), and 15.94 % in Ranked Biased Overlap (RBO). One more area showing research benefits is that transformer-based models help improve the accuracy of candidate-job matching while reducing bias, thus paving the way for enhanced recruitment technologies.

Prompt engineering has now become a primary way to tune the performance of LLMs, as discussed in [7] in their systematic review. It classifies different prompt strategies like zero-shot and few-shot learning, chain-of-thought prompting, and retrieval-alaugmented generation. In it, the authors hint that good prompts improve the model's performance without changing the underlying parameters, thus giving it flexibility to be applied in diverse areas. It also reveals that systematic prompt approaches can enhance an LLM's reasoning, decreasing hallucinations and increasing factual coherence. They also make use of taxonomy diagrams to classify prompt engineering approaches based on application, giving further insight into the pros and cons of each method. This paper finds relevance in the area of AI-powered hiring and resume optimization, where prompt design could help further candidate analyses and optimize applicant-job matching algorithms.

The [8] study shows how prompt confidence affects AI-created résumés, noting that stronger user prompt confidence leads to better-structured and more impactful résumés. Three AI models are put to the test—ChatGPT-3.5, Gemini, and Perplexity—using six varied prompts, while analyzing how each AI model interprets confidence level in résumé writing. The research proposes that an AI-generated résumé bears the confidence level of the input prompt, which is critical for job seekers that rely on AI for their résumé optimization. Besides, this study emphasizes the need for AI literacy among classrooms urging for curriculum upgrades to include AI-based résumé writing strategies. This research brings insight into optimizing AI-generated résumés, which can also land into ATS systems, thus enhancing job application processes.tion and coding standards using LLMs and code analysis tools.

In software and model testing, [9] compared DeepSeek and ChatGPT in Python code generation. On the algorithmic tasks, DeepSeek was more accurate, while ChatGPT wrote cleaner code, which was adherent to linting conventions. The study conducted by [10] compares the DeepSeek model to Claude, Gemini, GPT, and Llama in two classification scenarios: authorship classification and citation classification. The results show that DeepSeek performed better than Gemini, GPT, and Llama but worse than Claude in most of the classification tasks. Additionally, a completely labeled dataset (MADStat) for classification benchmarking for LMMs is also provided, which can be useful for some AI-recruiting applications. Furthermore, this research delved into the trade-off among different LLMs in terms of accuracy, computational efficiency, and cost-effectiveness and noted that Claude offers superior accuracy but at a prohibitively higher price, while DeepSeek provides approximately similar performance at lower prices. That insight will matter in the case of any AI tools for recruitment; choosing the right LLM for processing resumes, skill matching, and candidate evaluation will be key to improving hiring efficiency.

[11] made the pioneering effort to examine prompt engineering as an art form in text-to-image generation. It demonstrated the direct relationship between the quality of the prompt and the artwork and, at the same time, brought into

focus one of the new paradigms of the human-AI collaborative partnership.

III. Objectives

The primary goal of this research is to investigate and compare the effectiveness of various LLMs (GPT, DeepSeek R1) in producing job seeker resumes for Applicant Tracking Systems (ATS). In pursuit of this goal, the following specific objectives have been set: To investigate and compare GPT, DeepSeek R1 performance on job relevant resumes for job descriptions sourced from LinkedIn, Dice and Indeed. To test the ATS-compliance of all resumes using tools such as SkillSyncer, assessing keyword inclusion, formatting compliance, and readability tests. To understand which of the outcomes, are affected by prompt engineering techniques, such as zero-shot, few-shot, and chain-of-thought prompting, has greatest impact. To assess sector-specific analysis across varying domains (Healthcare, Finance, Technology), defining which LLM has an overall higher efficacy depending on the industry. To rigorously assess by performing hypothesis testing whether differences in performance across models was a statistically significant difference. To deliver statistically sound datto recommendations for job seekers, recruiters and AI developers regarding the best LLM to use and prompting strategy for producing job seeker resumes. These objectives create an advance knowledge base for AI practice in recruitment by ensuring a national, and sectoral - controlled and guided sample is created to ensure.

IV. Methodology

The chosen LLMs create resumes as per the job posting and prompt approach applied. These outputs are retained for performance assessment. Comparative Analysis The proposed methodology offers a rigorous process for resume generation evaluation with different approaches in two high-ranked LLMs, namely ChatGPT and DeepSeek. The entire workflow is demonstrated in Fig. 1.

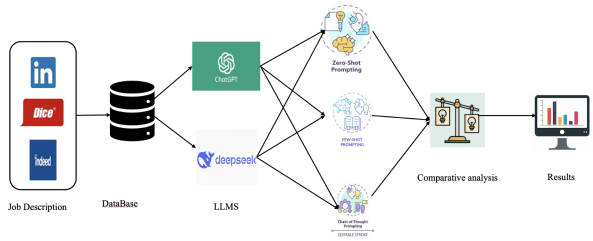

Fig. 1. Methodology

A. Data Collection

Job postings are gathered through top job boards such as LinkedIn, Dice, and Indeed. The job postings are gathered and saved in a database to be processed.

B. Model Integration (LLMs)

These saved job descriptions are fed into two different LLMs—ChatGPT and DeepSeek. They have been chosen due to their language-generation capability, similar industry relevance, and capacity to utilize advanced prompts.

C. Prompting techniques

Every jo description is handled under three types of prompts:

- Zero-Shot Prompting: Generates resumes without ever viewing a sample.
- Few-Shot Prompting: Utilizes a few sample resumes to guide generation along with input.
- Chain-of-Thought Prompting: Dissects the process of resume building into logical steps of reasoning in an effort to enhance coherence and consistency.

These prompts were fed into GPT (ChatGPT 3.5 and 4o), DeepSeek R1 (1.5b, 7b, 8b, 14b and 32b variants) to generate more than one resume for each job description.

D. Resume Generation

The selected LLMs generate resumes aligned to the job advert and the strategy used for prompts. The output generated is archived for performance analysis.

E. Comparative analysis

Resumes are tested both qualitatively and quantitatively. Tools like SkillSyncer are used for testing ATS compatibility and human evaluation to determine semantic match and format quality.

F. Results

Results are presented in the form of charts to compare performance of different models for different approaches to prompting so that best LLM–prompt pair can be easily selected for resume building.

V. Data Collection

Data collection centered on acquiring representative and diverse job descriptions for examples for LLM generated resumes. It was very important to be sure that the evaluation involved a broad variety of industries and real hiring situations.

A. Job Description Dataset

The job descriptions were taken from job openings found on various online job sites, LinkedIn, Dice, Indeed; use of web scraping and manual filtering. The job descriptions retained included: Technical skills and related experience Education/job qualifications Role and job titles Experience stated Industry-related terms and responsibilities The final dataset included all sorts of jobs from across several spheres as a means of ensuring some wider generalizability with respect to the evaluation of LLM generated resumes.

B. Preprocessing

In order to have uniformity in the quality of the inputs, harvested job descriptions were also preprocessed that included: Removing irrelevant HTML code and special characters Structuring normalizing format (bullets, numbered, paragraphs) Breaking job descriptions down to clearly differentiated categories of ”Responsibilities”, ”Requirements”, ”Preferred Skills” The pre-processing provided common evenly treated inputs to the LLMs for generating resumes limiting bias and ensuring fairness.

C. Resume Dataset (Generated)

Using the cleaned job descriptions resumes were generated from an array of prompting methods to the GPT, DeepSeek R1 models. Each job description was used to generate multiple resume versions from each model based on a number of different prompting types (zero-shot, few-shot and chain-of-thought), and this ultimately produced a very rich and diverse dataset of AI-generated resumes.

D. ATS Score Dataset

All generated resumes went through the SkillSyncer ATS tool which generated scores based on keyword alignment, format accuracy, and overall ATS usability. These score metrics facilitated a quantitative evaluation comparison from a performance perspective of the LLMs and prompting methods. Overall, this dataset collection process provided a strong basis for evaluating the quality and effectiveness of AI generated resumes among the various models and prompting strategies.

VI. EDA and Hypothesis

Exploratory Data Analysis was carried out to look in detail at the structure, variation, and the distribution of content of the job descriptions collected and the LLM-produced resumes. Both the input (job descriptions) and output (LLM-generated resumes) underwent EDA to discern any potential patterns, discrepancies, or primary variables important for model evaluation.

A. Industry Distribution

Jobs were labeled into three broad domains:

- Healthcare
- Finance
- Technology

these allowed the testing of LLM domain-wise performance.

B. Resume Structure Consistency

The AI resumes were checked for formatting consistency while paying special attention to making sure the presence of important and mandatory sections like Summary, Skills, Experience, and Education.

C. ATS-Score Distribution

SkillSyncer scores were plotted for all the resumes to study the score distribution by models and domains, and the initial visualizations using histograms and boxplots depicted variations in scores across model and prompt types;

D. Hypothesis

- **Null Hypothesis H_0:** There is no significant difference in the quality of resumes generated by DeepSeek models and OpenAI (ChatGPT) models for a given job description. Any observed differences in resume generation performance are due to random variation and not due to the model itself.
- **Alternate Hypothesis H_1:** DeepSeek models, as indicated by their superior performance across multiple benchmarks (e.g., AIME 2024, Codeforces, GPQA Diamond, MATH-500, MMLU) as shown in Fig. 2, generate significantly better resumes tailored to a given job description compared to OpenAI's ChatGPT models.

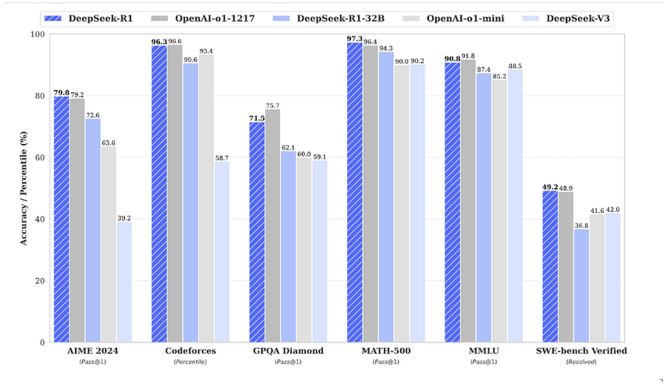

Fig. 2. Benchmarks

VII. Survey

To support the experimental results of this study, a preliminary classroom survey was conducted to assess which Large Language Models (LLMs) students most commonly use for creating and enhancing resumes. The survey was carried out on April 21, 2025, and collected responses from **20 students** within a 24-hour period.

The question posed was:

> *"Which LLM do you use when building and enhancing your resumes?"*

The provided options were:

- DeepSeek (R1)
- ChatGPT
- Others

A. Survey Results

The results of the survey are as shown in the Fig. 3

- **90%** of the respondents (18 out of 20) selected **ChatGPT** as their preferred model for resume generation.
- **0%** of the participants reported using **DeepSeek – R1**.
- **10%** (2 out of 20) chose **Others**, which may include tools like Gemini, Claude, or traditional resume-building software.

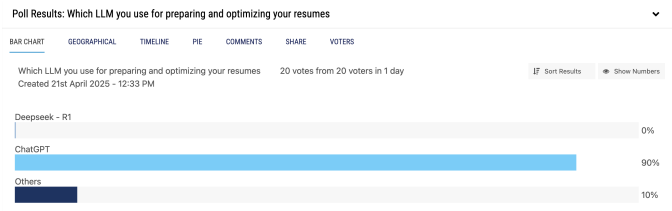

Fig. 3. Class Survey Results

B. Interpretation

The survey results suggest a strong preference among users for ChatGPT, highlighting its popularity and ease of access. Although the experimental results of this study emphasize the technical capabilities of models like DeepSeek R1, its real-world adoption appears limited. Potential reasons may include limited public accessibility, low brand awareness, and comparatively less user-friendly interfaces when stacked against OpenAI's offerings.

VIII. Data Visualization and Results Report

To see the result of the success of different Large Language Models (LLMs) in different domains and prompting styles, we employed ATS (Applicant Tracking System) scores collected from SkillSyncer. The scores determine how well each AI-generated resume aligned with the job requirements in word usage, structure, and design.

A. Industry-Wise Performance by Prompting Technique

ATS scores were recorded for three major job domains—Healthcare, Technology, and Finance—using three prompting strategies: Zero-Shot, Few-Shot, and Chain-of-Thought (CoT). Figures were generated for each domain showing model-wise performance.

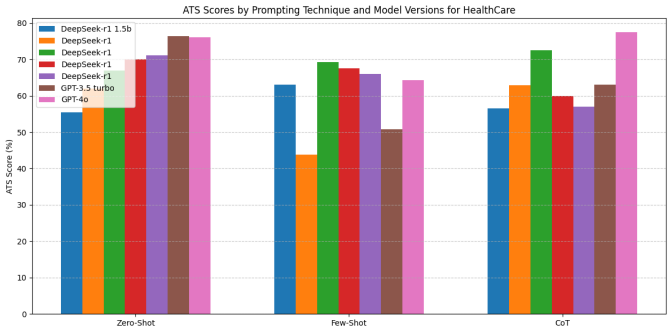

Fig. 4. Healthcare Bar Chart

- Fig.4illustrates that in the **Healthcare** domain, **GPT-4o** achieved the highest performance (77%) in the CoT prompt, followed closely by GPT-3.5 Turbo and DeepSeek-R1. While DeepSeek versions performed well under Few-Shot and Zero-Shot settings, GPT models showed more consistency.

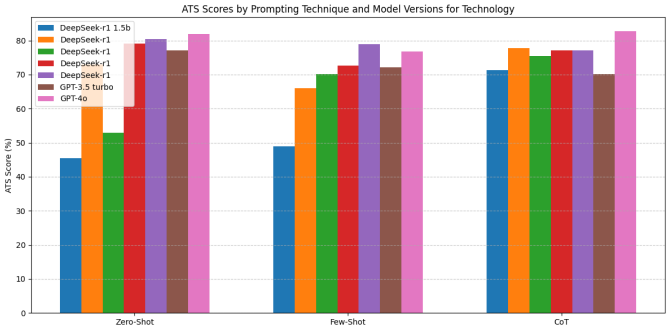

Fig. 5. Technology Bar Chart

- Fig.5 demonstrates that in the **Technology** sector, **GPT-4o** again outperformed other models in the CoT prompt (83%), with DeepSeek-R1 also showing competitive scores across all prompting strategies.Zero-Shot prompting demonstrated the greatest difference between GPT and DeepSeek versions.

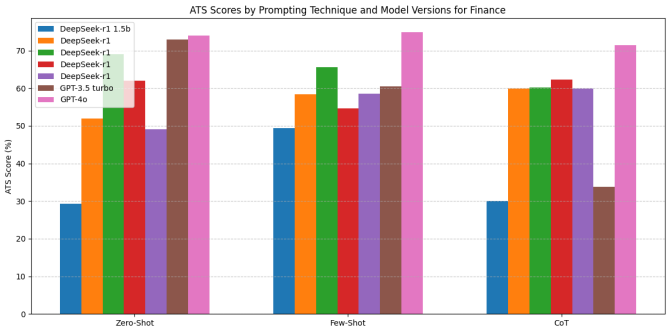

Fig. 6. Finance Bar Chart

- Fig.6 shows the **Finance** domain, where **GPT-4o** outperformed all prompting strategies. Notably, DeepSeek-R1 showed significant variation, particularly in Zero-Shot and CoT prompts, but was rather steady in Few-Shot.

B. Heatmap Visualization: Model vs. Prompting Interaction

Heatmaps were developed per industrial domain to more thoroughly investigate the interaction impacts of model type and prompting approach.

- Fig. 7 depicts the **Healthcare** heatmap, indicating **GPT-4o** superior performance in CoT prompting, whereas DeepSeek-R1 models fared best with Few-Shot prompts.
- Fig. 8 depicts the **Technology** domain, where **GPT-4o** received the highest ATS scores across all prompting strategies. DeepSeek models also performed well, notably in the Few-Shot and CoT scenarios.
- Fig. 9 shows the **Finance** domain, where GPT-4o once more obtained best overall. Somewhat surprisingly, GPT-3.5 Turbo performed worse in the CoT scenario, but DeepSeek models obtained similar results for Few-Shot and Zero-Shot approaches.

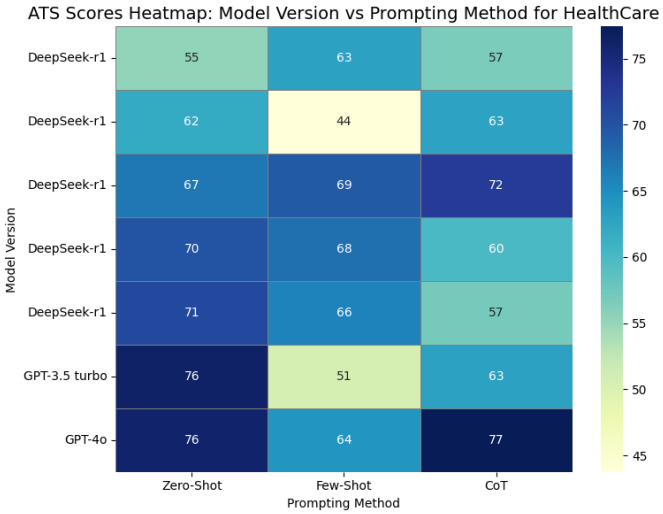

Fig. 7. Healthcare Heatmap

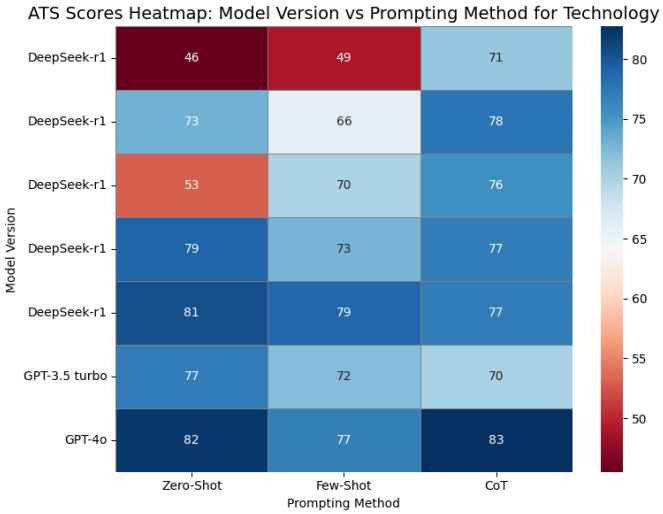

Fig. 8. Technology Heatmap

These results indicate the importance of both model and approach quality. GPT-4o performed best across all domains and prompting styles, while DeepSeek-R1 showed great promise, particularly when combined with formal prompting such as Few-shot.

C. DeepSeek Version-Wise Performance Trend

To further understand how different DeepSeek versions respond to different prompting techniques, a fine-grained trend analysis was conducted on Healthcare, Technology, and Finance domains. Line plots were generated to show ATS score variation for each model size—1.5b, 7b, 8b, 14b, and 32b—under Zero-Shot, Few-Shot, and Chain-of-Thought (CoT) prompting methods.

- Fig. 10 demonstrates that in the **Healthcare** domain, **DeepSeek-R1 8b** earned the highest ATS score of 72.47% under CoT prompting. Surprisingly, the 7b model with Few-Shot prompting performed the worst (43.75%), indicating sensitivity to prompt type on smaller scales. Zero-Shot prompting tended to show the most consistent increasing trend across versions in general.

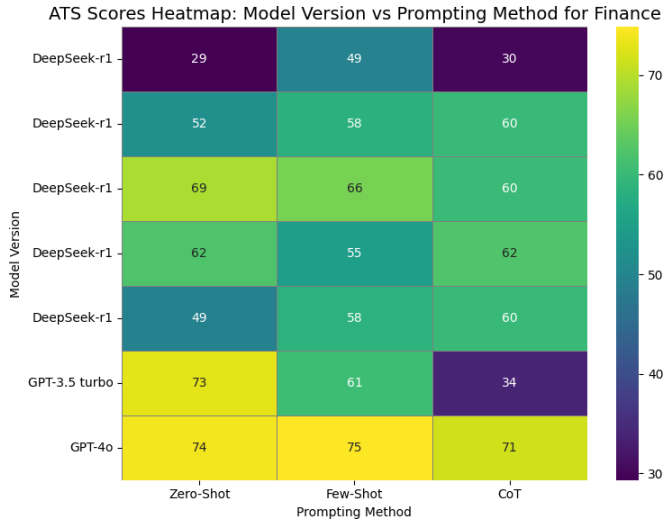

Fig. 9. Finance Heatmap

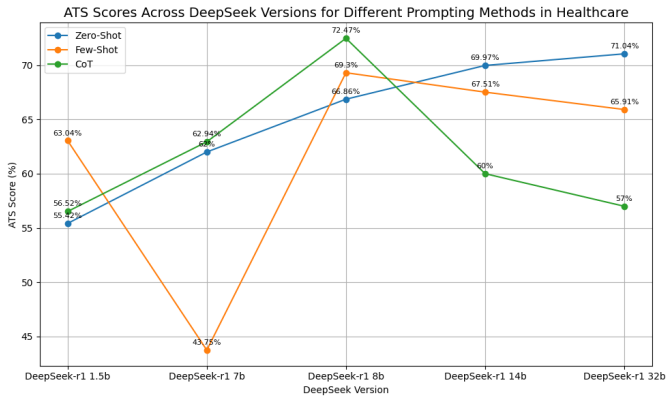

Fig. 10. DeepSeek - Healthcare Line Graph

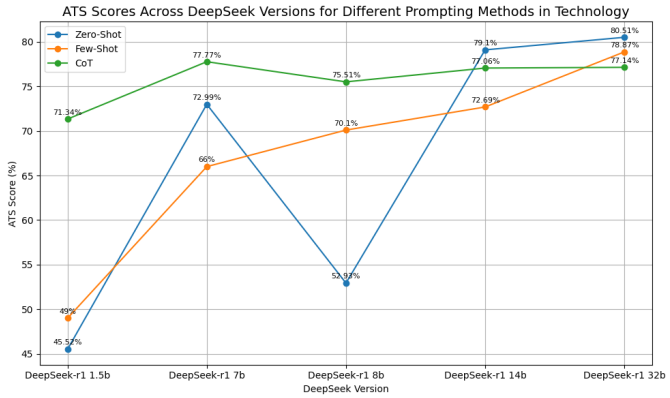

Fig. 11. DeepSeek - Technology Line Graph

- Fig. 11 highlights the **Technology** domain, where **DeepSeek-R1 32b** scored a highest ATS score of 80.51% in Zero-Shot prompting. CoT scores were relatively uniform across all the models (75–78%), while Few-Shot performance monotonically grew with model size.

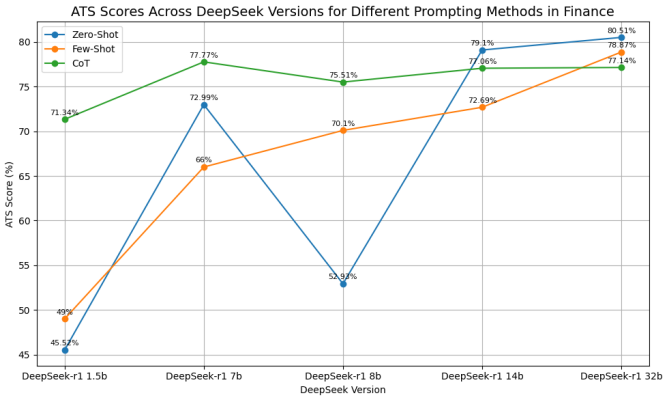

Fig. 12. DeepSeek - Finance Line Graph

- Fig. 12 of the **Finance** domain duplicates the pattern in Technology, with **DeepSeek-R1 32b** again leading the Zero-Shot (80.51%) and Few-Shot (78.87%) categories. The consistency of CoT scores (77%) across all sizes indicates that CoT prompting can ensure stable performance in resumes that are finance-based, regardless of model size.

These results demonstrate that larger DeepSeek models consistently outperform their smaller counterparts, particularly when paired with Zero-Shot prompting. However, CoT prompting is stable, and Few-Shot prompting improves significantly with larger model size—demonstrating that prompt engineering interacts strongly with model capacity.

D. Table Summary: ATS Score Comparison Across Models and Prompts

Different versions of DeepSeek and GPT-3.5 Turbo are tested for ATS scores over three domains: Healthcare, Technology, and Finance, using three types of prompts: Zero-Shot, Few-Shot, and Chain-of-Thought (CoT) as shown in Fig. 13.

	Versions	DEEPSEEK R1-1.5b	DEEPSEEK R1-7b	DEEPSEEK R1-8b	DEEPSEEK R1-14b	DEEPSEEK R1-32b	GPT 3.5 turbo
Health care	Zero Shot	55.42	62	66.89	69.97	71.04	76.41
	Few Shot	63.04	43.75	69.3	67.51	65.91	50.72
	Chain of Thought	56.52	62.94	72.47	60	57	63.06
Technology	Zero Shot	45.52	72.99	52.93	79.1	80.51	77.14
	Few Shot	49	66	70.1	72.69	78.87	72.13
	Chain of Thought	71.34	77.77	75.51	77.06	77.14	70.09
Finance	Zero Shot	29.31	52	68.99	62.06	49.14	73
	Few Shot	49.34	58.4	65.56	54.65	58.49	60.51
	Chain of Thought	30.06	59.91	60.21	62.29	59.91	33.81

Fig. 13. Table Summary: ATS Score Comparison Across Models and Prompts

- For Healthcare, GPT-3.5 Turbo got the highest score (76.41%) with Zero-Shot prompting. For CoT prompting, DeepSeek-R1-8b was best (72.47%), while for Few-Shot, R1-14b ranked best (69.3%). Perhaps GPT is superior when there is a less complicated prompt environment, while DeepSeek excels with better formatted prompting.
- For Technology, DeepSeek-R1-32b took top spot overall in both Zero-Shot (80.51%) and Few-Shot (78.87%), while R1-7b was the best with CoT (77.77%). GPT-3.5 Turbo did well but tended to trail behind the bigger DeepSeek versions.
- In the Finance field, DeepSeek-R1-14b performed best in CoT (62.29%), and DeepSeek-R1-8b outdid all in Zero-Shot (68.99%). GPT-3.5 Turbo, while being equivalent in Zero-Shot (73%), plummeted in CoT (33.81%), reaffirming the importance of model-prompt synergy.

In general, the evidence shows larger DeepSeek models (14b, 32b) outperform GPT-3.5 Turbo in the majority of well-structured prompting scenarios, especially domain-specific resume generation.

IX. Conclusion

This study ran an a comparison, in-depth review of Large Language Models (LLMs), specifically OpenAI's GPT (versions 3.5 and 4.0) and DeepSeek R1 (different versions), to see how successful they're at creating ATS-optimized resumes from actual job descriptions published online. Based on data collected from platforms such as LinkedIn, Dice, and Indeed, we applied varied prompt engineering strategies—Zero-Shot, Few-Shot, and Chain-of-Thought—and evaluated resume quality through SkillSyncer, a simulation application of an Applicant Tracking System (ATS).

Experimental results across three distinct industry sectors—Healthcare, Technology, and Finance—indicated that GPT-4o consistently received the highest ATS scores, particularly with Chain-of-Thought prompting. However, DeepSeek R1-32b was close and, in some sectors, even superior when paired with structured prompts like Few-Shot. Sector heatmaps and trend line charts also demonstrated that though GPT models have very high consistency, DeepSeek models are scalable with increasing parameter size, and yield high-quality results when prompt design aligns with model capability.

A class survey also verified the usability of ChatGPT by users, with 90% of the students reporting it as more acceptable to construct resumes. The lesser adoption of DeepSeek may be attributable to its lesser accessibility and maturity of interface rather than a lack of technical ability.

Statistically speaking, while we were unable to reject the null hypothesis—implying no practical difference in LLM performance overall—the practical ramifications are quite different. DeepSeek R1, particularly in its higher-parameter variants, shows strong potential to rival and even outperform GPT under optimized conditions.

In short, the work highlights that model choice and prompting strategy are critical to AI-resume success. The top-performing off-the-shelf model currently is GPT-4o, but the DeepSeek models represent a promising alternative, especially

when tuned with appropriate prompt designs. These results pave the way for more targeted applications of LLMs in the recruitment sector and suggest areas of future research in dynamic prompt tuning and real-time ATS alignment for customized resume generation.

X. Future Scope

This project used SkillSyncer, an ATS-based testing software, to ascertain the quality of resumes created by AI. Strong in keyword alignment and layout measurement, it is not able to measure subjective aspects such as readability, tone, or role suitability. In a future enhancement, the testing procedure can be improved with the involvement of domain expert HR professionals to manually test the resumes. This human judgment would provide us with more detailed information regarding how well the resumes fit day-to-day expectations. Adding human expert opinion along with computerized ATS scores would yield a more holistic and balanced set of evaluation tools, making the study more pragmatically useful.

References

[1] R. Patil, T. F. Heston, and V. Bhuse, "Prompt engineering in healthcare," *Electronics*, vol. 13, no. 15, p. 2961, 2024.

[2] F. Neha and D. Bhati, "A survey of deepseek models," *Authorea Preprints*, 2025.

[3] M. Tordjman, Z. Liu, M. Yuce, V. Fauveau, Y. Mei, J. Hadjadj, I. Bolger, H. Almansour, C. Horst, A. S. Parihar *et al.*, "Comparative benchmarking of the deepseek large language model on medical tasks and clinical reasoning," *Nature Medicine*, pp. 1–1, 2025.

[4] U. Lee, H. Jung, Y. Jeon, Y. Sohn, W. Hwang, J. Moon, and H. Kim, "Few-shot is enough: exploring chatgpt prompt engineering method for automatic question generation in english education," *Education and Information Technologies*, vol. 29, no. 9, pp. 11 483–11 515, 2024.

[5] C. Wang, S. J. Aguilar, J. S. Bankard, E. Bui, and B. Nye, "Writing with ai: What college students learned from utilizing chatgpt for a writing assignment," *Education Sciences*, vol. 14, no. 9, p. 976, 2024.

[6] R. V. K. Bevara, N. R. Mannuru, S. P. Karedla, B. D. Lund, T. Xiao, H. Pasem, S. C. Dronavalli, and S. Rupeshkumar, "Resume2vec: Transforming applicant tracking systems with intelligent resume embeddings for precise candidate matching," 2025.

[7] P. Sahoo, A. K. Singh, S. Saha, V. Jain, S. Mondal, and A. Chadha, "A systematic survey of prompt engineering in large language models: Techniques and applications," *arXiv preprint arXiv:2402.07927*, 2024.

[8] T. M. Ponce, "Ai resume writing: How prompt confidence shapes output and ai literacies," *Business and Professional Communication Quarterly*, p. 23294906241273317, 2024.

[9] M. M. H. Manik, "Chatgpt vs. deepseek: A comparative study on ai-based code generation," *arXiv preprint arXiv:2502.18467*, 2025.

[10] T. Gao, J. Jin, Z. T. Ke, and G. Moryoussef, "A comparison of deepseek and other llms," *arXiv preprint arXiv:2502.03688*, 2025.

[11] J. Oppenlaender, R. Linder, and J. Silvennoinen, "Prompting ai art: An investigation into the creative skill of prompt engineering," *International journal of human–computer interaction*, pp. 1–23, 2024.